

Agents That Are Toxic to Dairy Cattle

INTRODUCTION

A variety of naturally occurring toxic agents can be present in harvested, grazed, or purchased feedstuffs consumed by dairy cattle. These include compounds produced by plants, fungi, and microbes naturally occurring in feeds and by microbes that may contaminate feed or water. Through ingestion of an adequate dose, these agents can cause dysfunction, sickness, or death to the animal and do not have a recognized nutritional value at other dosage levels. Some ingested toxic agents may pass into milk or animal tissues, which then may enter the human food supply. With the intent of preserving human health, this contamination of ruminant-derived foodstuffs is the basis for regulatory limits on levels in animal feeds imposed by the U.S. Food and Drug Administration (FDA). The major, naturally occurring toxic agents likely to be consumed by dairy cattle and are of interest for human health are addressed in this chapter. The effects of excess minerals are briefly discussed in Chapter 7 of this book, and an extended discussion can be found in Mineral Tolerance of Animals (NRC, 2005). Manufactured toxins, such as pesticides and herbicides, and toxic rangeland plants are not addressed.

This chapter contains a general overview of individual toxic agents, their potential to be transmitted to milk, and general management practices for prevention of toxicoses.¹

¹For regulatory issues related to toxins, the action levels and specification as to whether they are expressed on an as fed or dry matter basis should be verified with current U.S. Food and Drug Administration (FDA) regulatory guidance. Further information on toxic agents may be found at the FDA poisonous plant database at www.accessdata.fda.gov/scripts/planttox/index.cfm; the FDA Mycotoxin Regulatory Guidance at www.ngfa.org/wp-content/uploads/NGFAComplianceGuide-FDARegulatoryGuidanceforMycotoxins8-2011.pdf; the Merck Veterinary Manual, 10th ed., 2010, C. M. Kahn, ed. Merck & Co, Inc., Whitehouse Station, NJ, at www.merckvetmanual.com/mvm/index.html; and FDA. 2012. Bad Bug Book, Foodborne Pathogenic Microorganisms and Natural Toxins, 2nd ed., Lampel, K. A., ed. www.fda.gov/downloads/Food/FoodborneIllnessContaminants/UCM297627.pdf.

Readers should seek the advice of veterinarians for treatment of specific toxicosis issues.

PLANT COMPOUNDS AND AGENTS PRODUCED IN PLANTS

Plants and the fungi that live within them (endophytes) can produce and accumulate compounds that may be harmful to animals that consume them. Many of these compounds are part of the natural defense mechanisms of the plants. Their production can vary with environmental conditions, season, plant part, and cultivar.

Alkaloids in Feeds

Alkaloids are produced in cool-season forage grasses, including *Phalaris* spp. (e.g., reed canarygrass = *Phalaris arundinacea*) and *Lolium* spp. (e.g., perennial ryegrass = *Lolium perenne*; tall fescue = *Lolium arundinaceum*) (Cheeke, 1995). Sprouted and sunburned green potatoes may also contain toxic alkaloids. The alkaloids produced in each grass are diverse, although within a grass, a specific alkaloid may be responsible for much of the negative effect. The alkaloids may be intrinsic to the grass or produced by endophytes associated with the grass (Cheeke, 1995).

Phalaris spp. may contain at least eight different tryptamine and P-carboline alkaloids, some of which have structural similarity to the neurotransmitter serotonin (Cheeke, 1995). *Phalaris* staggers is a neurologic condition of cattle and sheep related to the consumption of *Phalaris* spp. It may be acute and reversible or a chronic, irreversible, and typically lethal form (Binder et al., 2010). Animals may suffer from a progression of symptoms of staggering, ataxia, recumbency, and death; symptoms may be delayed by 1 month or more from the time of *Phalaris* ingestion (Binder et al., 2010). Even in the absence of such symptoms, the performance of animals consuming *Phalaris* spp. is less than what would be expected from the composition of the grass (Cheeke, 1995).

Lambs fed low-alkaloid cultivars of reed canarygrass had similar dry matter intakes (DMIs) and dry matter (DM) and protein digestibilities as those fed timothy (Tosi and Wittenberg, 1993). Ensiling reed canarygrass did not alter alkaloid content of the forage (Tosi and Wittenberg, 1993).

Production of alkaloids in tall fescue and perennial ryegrass is associated with the presence of endophytes (*Neotyphodium coenophialum* and *Neotyphodium lolii*, respectively) (Canty et al., 2014). The major alkaloids in endophyte-infested tall fescue are lolines and ergovaline, with the latter, an ergot alkaloid with vasoconstrictive effects, being most detrimental to animal health (Porter, 1995). However, ergovaline toxicosis is likely augmented by other alkaloids in the infected grass (Porter, 1995). Symptoms of tall fescue toxicosis in cattle include lameness, body weight (BW) loss, dull appearance, rough haircoat, dry gangrene of the extremities, elevated body temperatures, elevated respiration rates, altered hoof growth (Jacobson et al., 1963), fat necrosis (Stuedemann et al., 1985), and reduction in serum prolactin concentrations (Hurley et al., 1981). Responses to the toxins are affected by environmental temperature (e.g., elevated body temperatures and respiration rates in warmer weather, increased lameness, and dry gangrene in cold weather) (Jacobson et al., 1963), but toxins are present throughout the growing season (Hemken et al., 1981). Animal toxicosis has been reported to occur at 50 ng ergovaline/g of tall fescue grass (Porter, 1995). Replacement of endophyte-infected forage with low-endophyte or endophyte-free tall fescue or other grasses, interseeding legume forages, or other approaches to diluting the potential dose of alkaloids usually reduce the risk of tall fescue toxicosis.

Perennial ryegrass contains at least two classes of toxins: lolitrem alkaloids and ergopeptide alkaloids (Hovermale and Craig, 2001). Tremorgens, including lolitrem B, are potent neurotoxins that cause ryegrass staggers. This toxicosis is characterized by incoordination, staggering, head shaking, and collapse, although death is not a direct consequence of the intoxication (Cheeke, 1995). Lolitrem B concentrations are greatest in the basal halves of leaf sheaths, lowest in the leaf blades, and greater in the spring season (di Menna et al., 1992). Because the site of greatest lolitrem B concentrations in the plant is in the basal portion of the leaf sheath, the staggers syndrome is most often seen in closely grazed pastures (Cheeke, 1995). Ryegrass staggers can occur at 5 µg of lolitrem B/g of grass (Porter, 1995). Recovery is spontaneous in 1 to 2 weeks if animals are fed nontoxic pastures or feeds (Merck, 2010c).

Cows fed endophyte-infected perennial ryegrass containing a maximum daily dose of approximately 35 mg lolitrem B produced milk with a maximum of 5 ng lolitrem B/mL milk (Finch et al., 2013). Lolitrem B concentrations in milk returned to almost zero 8 days after withdrawal of lolitrem B-containing forage. References were not found for dairy cattle describing the transfer to milk of alkaloids associated with *Phalaris* spp. or other *Lolium* spp. A study evaluating

effects of endophyte-infected tall fescue on ewes was unable to detect ergovaline in the milk at a limit of detection of 0.15 ng/mL when animals were fed diets averaging 497 pg ergovaline/kg of feed DM (Zbib et al., 2014). Further work is needed to determine carryover into milk of plant alkaloids and factors that may affect it.

Bracken Fern

Bracken fern (*Pteridium aquilinum*) is broadly distributed in the United States and contains a variety of toxins, including ptaquiloside (Potter and Baird, 2000). Poisoning may occur through consumption of bracken fern in pasture or hay. Consumption of the fern by cattle can result in the presence of blood in the urine, extensive hemorrhages throughout the body, aplastic anemia, and development of tumors (papillomas and carcinomas) in the bladder (Langham, 1957; Pamukcu et al., 1976). Extracts of the plant have mutagenic and carcinogenic properties and can pass through into milk to affect suckling calves (Evans et al., 1972) and cause cancers in milk-fed mice (Pamukcu et al., 1978). One report indicated that ptaquiloside is excreted in milk in a linear, dose-dependent fashion. The concentration in milk was approximately 8.6 ± 1.2 percent of the amount ingested by the cow (Alonso-Amelot et al., 1996). Ingestion of bracken fern toxins has been associated with various cancers and health disorders in humans (Shahin et al., 1999). Bracken fern also contains thiaminases, but these are less likely to affect ruminants than nonruminants due to the production of thiamin in the rumen (Merck, 2010c). Preventing cattle from consuming bracken fern through removal of the fern from pastures and hay fields or providing ample nonfern forage in pastures is strongly recommended.

Coumarin

Coumarin is a polyphenolic compound naturally occurring in many plant materials, including sweet clover forage (*Melilotus alba*, *Melilotus officinalis*). When coumarin is oxidized to dicumarol (3,3'-methylene-bis[4-hydroxycoumarin] or bishydroxycoumarin), it becomes a powerful anticoagulant (Stahmann et al., 1941). Molds convert coumarin to dicumarol, which interferes with synthesis of vitamin K coagulation factors and prothrombin (Radostits et al., 1980). "Sweet clover disease" is a hemorrhagic disease of cattle and sheep that have consumed moldy or spoiled sweet clover forage in which the blood ceases to clot (Roderick, 1931). The hemorrhaging may be visible, such as substantial bleeding that does not stop after injury or surgery, or may not be visible if the bleeding is internal. Consumption of hay with as-fed dicumarol concentrations of 20 to 30 mg/kg over several weeks may cause poisoning and death in cattle (Merck, 2010c). Calves born from cows consuming forage containing dicumarol also suffer from the hemorrhagic disorder (Roderick, 1931) due to transfer of the toxic agent across the

placenta during pregnancy (Merck, 2010c). In time, animals can recover if feed free of dicumarol is provided. Information regarding passage of dicumarol to milk was not found.

Gallotannin

Gallotannin is a hydrolysable tannin found in blossoms, buds, young leaves, and acorns of oaks (*Quercus* spp.) that can be poisonous to cattle. Gallotannins hydrolyze to lower molecular weight compounds that denature cell proteins, resulting in cell death and then in necrosis and lesions in the gastrointestinal tract, liver, and kidneys. Clinical signs of “oak poisoning” of cattle occur in 3 to 7 days from the time of ingestion and include anorexia, depression, brisket edema, rumen atony, and constipation changing to bloody diarrhea (Merck, 2010c). Blood urea N and creatinine may be elevated, consistent with renal failure, and ulceration or necrosis of the ruminal, omasal, and abomasal linings is present (Perez et al., 2011). If animals consume other feeds after ceasing consumption of oak materials, they generally recover completely. Prevention of the consumption of gallotannin-containing material is recommended. Management strategies to accomplish this include maintaining adequate forage access when cattle graze, keeping cattle off pastures with oak trees until trees are mature, and closely observing animals for acorn consumption so they may be removed from the pasture if this occurs. Gallotannins have also been found in the leaves of red maple (*Acer rubrum*) (Agrawal et al., 2013), but no reports were found associating these with toxicoses in cattle. Information was not found on passage of the toxic principles into milk.

Glucosinolate Goitrogens

These are a class of naturally occurring, sulfur-containing compounds produced by plants of the Brassica spp. such as rape (*Brassica campestris*) and mustard (*Brassica* spp.) that negatively affect iodine metabolism. See the Iodine section in Chapter 7 for details.

Gossypol

This naturally occurring, toxic, yellow pigment is found in cottonseed (*Gossypium* spp.) and cottonseed products, including cottonseed meal. Free rather than bound gossypol is the biologically active form (Randel et al., 1992). The content of free gossypol in cottonseed kernels (*Gossypium hirsutum*) varies from 0.59 to 2.35 percent of DM, with an average of 1.32 percent and standard deviation of 0.35 percent (Pandey and Thejappa, 1975). This is approximately equivalent to about 0.9 percent of the weight of whole cottonseed on an as-received basis. Efforts have been devoted to genetically selecting cotton varieties with lower levels of gossypol. Although the rumen microbes can detoxify gossypol, feeding excessively high levels of free gossypol to cows

(Lindsey et al., 1980) or to animals without functionally developed rumens (Risco et al., 1992) can result in sickness or death of the animal. For lactating dairy cows, no detrimental effects on DMI or on lactation performance were observed with diets containing cottonseed or cottonseed meal providing 0 to 1,050 mg free gossypol/kg diet DM (Mena et al., 2001). However, cows in that study showed increased erythrocyte fragility with free gossypol at the 1,050 mg/kg level. Diets with up to 200 mg/kg of DM of free gossypol had no deleterious effects on calves through 90 days of feeding, but provision of 400 mg/kg or more free gossypol was toxic (Risco et al., 1992). Accordingly, feeding cottonseed products to young calves is not recommended. Gossypol can also affect reproductive function of bulls. Compared to control animals, yearling bulls consuming cottonseed meal (7 percent) and hulls (18 percent) or whole cottonseed (15 percent) and cottonseed hulls (17 percent, values as percentage of diet DM) for 2 months showed histological changes indicative of damage to the spermatogenic tissues and associated cells (Arshami and Ruttle, 1988). The damage was partially reversed after animals were fed a gossypol-free diet for 2 months. Reproductive function in cows is relatively insensitive to gossypol (Randel et al., 1992).

Gossypol was not detected in the milk of cows consuming diets containing 1,510 mg free gossypol/kg DMI, even when the animals showed negative responses related to gossypol toxicity (Lindsey et al., 1980). The sensitivity of the gossypol assay used in that study was 0.5 pg gossypol/mL of milk. In a later study, lactating dairy cows consuming 10 and 15 percent of diet DM as whole cottonseed providing 817 and 1,295 free gossypol mg/kg diet DM, respectively, had milk gossypol concentrations of 0.13 and 0.22 mg gossypol/kg milk after consuming the diets for 60 days (Wang et al., 2012). No gossypol was detected in the milk of cows consuming the control diet that contained no cottonseed product or the diets containing cottonseed meal and that provided no more than 117 mg free gossypol/kg diet DM. The allowable amount of free gossypol in cottonseed products added to food intended for human consumption is 450 mg/kg (FDA, 2015d).

Nitrate

Nitrate (NO_3^-) is a normal component of the vegetative portions of plants and may also be present in ground water. Concentrations can increase to toxic levels in plants grown under conditions of drought, reduced light intensity (Reid and Jung, 1980), and increasing nitrogen (N) fertilization (Murphy and Smith, 1967). Different plant species differ in the degree to which they accumulate NO_3^- , with sudangrass (*Sorghum sudanense*), orchardgrass (*Dactylis glomerata*), and tall fescue (*Festuca arundinacea*) accumulating NO_3^- to the greatest extent and alfalfa (*Medicago sativa*) and wheat (*Triticum aestivum*) the least (Murphy and Smith, 1967). Corn (*Zea mays*) was not evaluated in that study, but it can

accumulate potentially toxic levels of NO_3^- . NO_3^- toxicity occurs when excessive amounts of NO_3^- are converted to nitrite (NO_2^-) in the rumen and then not completely converted to ammonia (Carlson and Breeze, 1984). The NO_2^- is absorbed into the blood and converts hemoglobin to methemoglobin, which cannot transport oxygen (Reid and Jung, 1980), and turns the blood chocolate brown in color. In addition, when high NO_3^- forage is ensiled, NO_3^- can be converted to nitro-gen dioxide (NO_2), the toxic component in silo gas (Wang and Burris, 1960), which can cause lung damage or death to people or animals that inhale the gas. The amount of NO_2^- that can be toxic is variable, depending on the concentration in the plant, the rate at which the plants are consumed, the adaptation of the animal to the higher NO_3^- feed, and what other feeds are provided (Merck, 2010c). Early signs of toxicosis include subnormal body temperature, muscular tremor, weakness, and ataxia; brown-tinted mucous membranes develop as the amount of methemoglobin in the blood increases (Merck, 2010c). Consult Cooperative Extension bulletins (e.g., Strickland et al., 2011) for specific information on recommended forage and animal management practices for avoiding NO_2^- toxicity in livestock. Feeding NO_3^- can reduce enteric methane emissions; risk of toxicity due to increased NO_3^- consumption can be reduced through gradual acclimation of animals to dietary NO_3^- (Lee and Beauchemin, 2014).

Data were not found on NO_3^- or NO_2^- concentrations in milk from animals consuming naturally occurring elevated levels of NO_3^- . However, in a study in which up to 150 g of potassium nitrate (KNO_3 ; 92 g NO_3^-) was provided to lactating cows as a single, liquid oral dose, milk NO_3^- concentration increased to a maximum of 35.6 mg NO_3^- /L (standard deviation, 10.1 mg/L) and then returned to predose levels by 50 hours (Baranova et al., 1993). Level of milk production, diets, and feed intakes were not specified for the study. NO_2^- and NO_3^- concentrations in commercially available milk (2 percent milk fat) were 0.20 and 0.0002 mg/100 mL of milk, respectively (Hord et al., 2011). For comparison, human milk in that study contained 0.31 and 0.001 mg/100 mL for NO_2^- and NO_3^- , respectively. The

FDA limits NO_3^- measured as N to 10 mg/L bottled water (equivalent to 44 mg NO_3^- /L) (FDA, 2015b) or NO_3^- as sodium nitrate (NaNO_3) used as a preservative in cured or preserved fish or meat products to not more than 500 mg/kg (364 mg NO_3^- /kg product) (FDA, 2015c).

Prussic Acid

Prussic acid is another name for hydrocyanic acid (HCN). Cyanide poisoning of livestock occurs when the cyanogenic glycosides held in vacuoles within plant cells come in contact with plant or microbial enzymes capable of cleaving off and releasing HCN (Merck, 2010c). The HCN blocks the action of cytochrome oxidase in cellular respiration (Kingsbury, 1958). The total amount of glycoside and free HCN in the plant as well as the rate of ingestion of the toxic plants and

the size of the animal are important to determining whether poisoning will occur (Kingsbury, 1958). One study reported a minimum lethal dose of about 4.4 mg of HCN/kg of BW per hour, with death following ingestion of a lethal dose within 15 minutes to a few hours (Kingsbury, 1958). Alternately, approximately 2 mg of HCN/kg of BW may be a toxic dose for most animals (Merck, 2010c).

Prussic acid is found in vegetative matter from Sorghum spp. (Johnsongrass, sorghums, sudangrass), oats, wheat, rye, ryegrass, millet, chokecherry, and wild cherry, among many other sources. The Sorghum spp. are of greatest concern, particularly when grazed, because intakes can be high. A generalized ranking for potential HCN accumulation in sorghum types is Johnsongrass > sorghums > sorghum-sudan hybrids > sudangrass (Strickland et al., 2014). The potential HCN content of sorghum forage varies by variety and decreases with increasing maturity past 45 days (Gorashi et al., 1980) but increases with N fertilization (McBee and Miller, 1980). Amounts of HCN and risks of poisoning are likely to increase under stress conditions, such as drought or freeze damage, and are generally highest in young growing plants (i.e., short plants), the youngest material in older plants, and leaves as compared to stems (Strickland et al., 2014). Consult Cooperative Extension bulletins (e.g., Strickland et al., 2014) for specific information on recommended management practices for avoiding prussic acid poisoning of cattle grazing Sorghum spp.

HCN was shown to pass into milk in goats by virtue of increased blood HCN values in kids that suckled dams dosed with 1, 2, or 3 mg of potassium cyanide/kg of BW per day (Soto-Bianco and Gorniak, 2003). Information regarding the degree of transfer of HCN into milk was not found.

Trypsin Inhibitor

This globulin protein found in raw soybeans irreversibly binds with trypsin, inhibiting the action of the small intestinal protease (Kunitz, 1947). The inhibition of trypsin reduces protein digestibility in the small intestine. Heat treatment of soybeans or soybean meal denatures the inhibitor (Rackis, 1974), and most commercially available soybean meal is heat treated. Although typically posing no major issues to mature ruminants because the inhibitor is degraded and inactivated in the rumen (Hoffmann et al., 2003), feeding raw soybeans to young calves reduces protein digestion and impairs performance.

Another naturally occurring trypsin inhibitor is found in bovine colostrum; trypsin inhibition activity in colostrum appears to decline with day postcalving (Laskowski and Laskowski, 1951). In contrast to the deleterious effects of raw soybean meal fed to young calves, addition of soybean trypsin inhibitor to colostrum in the first two feedings increased serum concentrations of immunoglobulin G (IgG) and immunoglobulin M (IgM) in neonatal Jersey calves,

suggesting that the treatment improved transfer of passive immunity (Quigley et al., 1995). It seems likely that the natural presence of trypsin inhibitor in colostrum effectively reduces the degradation of immunoglobulins.

MYCOTOXINS

Mycotoxins are naturally occurring toxins produced by molds. Their prevalence in plant-derived feeds depends on growing conditions, damage to the plant, moisture/humidity and availability of oxygen during storage, concentration or dilution during processing, and so on. Mycotoxins may be present in the variety of feeds provided to cattle, including silages, grains, pasture, hays, and by-product feeds, and can impair animal performance. In addition to direct effects on the animal, some mycotoxins may have antibiotic properties that can affect rumen microbiota (Gallo et al., 2015) and so may have an indirect impact on performance. Because the molds and their toxins may not be evenly mixed throughout feed sources, appropriate sampling to determine the presence or absence of toxins can be a challenge. For mycotoxins that are regulated in feeds, the FDA mandates the analysis of feeds be performed by methods found in “(1) the most recent edition of the Official Methods of Analysis of the AOAC, (2) the FDA Laboratory Information Bulletins, or (3) peer reviewed literature” (FDA, 2005a). Advanced detection technologies such as liquid chromatography coupled to mass spectroscopy are allowing detection of more of these fungal metabolites, laying the basis for future investigations on their impact (Gruber-Dominger et al., 2017).

Aflatoxin

Aflatoxins are a group of toxins (B_1 , B_2 , G_1 , G_2) produced by the molds *Aspergillus flavus* and *Aspergillus parasiticus* (FDA, 2012a). They can be found in corn, sorghum, rice, cottonseed, peanuts, and a variety of other food crops (FDA, 2012a). Climatologically, the production of aflatoxins is associated with above-average temperatures and below-average rainfall during the growing season. Toxins may be produced while crops are in the field or in storage if moisture content and temperatures support mold growth (Merck, 2010c). Aflatoxins can cause health disorders in animals, with calves being very susceptible to their effects and adult ruminants being relatively resistant (Merck, 2010c). Exposure to aflatoxins can cause impaired liver function and reduced feed intake (Fink-Gremmels, 2008). Aflatoxins are potential carcinogens.

The greatest concern with aflatoxins is their potential impact on human health. Unlike some other mycotoxins, aflatoxin B_1 is not modified after incubation with rumen fluid (Kiessling et al., 1984), but it is converted to aflatoxin M_1 in the liver, and M_1 can pass into the milk. An estimated rate of aflatoxin B_1 as aflatoxin M_1 carryover into milk is 2.0 to 6.2 percent of intake (Fink-Gremmels, 2008). Although aflatoxin M_1 is “by far, not as hazardous as the parent compound,”

the FDA limits the allowable levels in milk to 0.5 pg/kg, primarily because milk tends to provide a substantial part of the diets of infants and children (FDA, 2012a). To reduce human exposure to aflatoxins in animal products, limits are placed on the concentrations of aflatoxins allowable in the diets of different classes of cattle. Current FDA regulations for dairy animals limit aflatoxin to no more than 20 pg/kg for corn, peanut products, cottonseed meal, and other animal feeds and feed ingredients (FDA, 1994). For dairy animals raised as beef cattle, corn or peanut products intended for finishing animals may contain 300 pg/kg aflatoxin; cottonseed meal intended for beef cattle may contain 300 pg/kg aflatoxin regardless of age (FDA, 1994). The allowable aflatoxin values are likely on an approximately 88 percent DM basis to make them equivalent to as fed dry com grain, cottonseed, and peanut products.

Deoxynivalenol

Deoxynivalenol (DON; vomitoxin) is a trichothecene mycotoxin produced by *Fusarium* spp. and other mold species. Feeds that contain DON also typically contain zearalenone, another *Fusarium*-produced mycotoxin (Mirocha et al., 1976). Among potential sources, wheat can be a major source of DON, but contamination has been reported for all major grain commodities (Price et al., 1993; Bianchini et al., 2015). Forages may also contain DON (Keller et al., 2013). Deoxynivalenol can reduce the synthesis of proteins in affected animals, but the main apparent effect is usually reduced feed intake (Pestka, 2007). Cattle, however, appear to be largely resistant to the negative effects of DON on feed intake and milk production (Côté et al., 1986), possibly because ruminal and intestinal microbes convert DON to deoxy DON (DOM-1) (King et al., 1984; Pestka, 2007). In a bioassay for toxicity, DOM-1 did not inhibit growth of yeast cells as compared to DON, which inhibited growth at 23 mg/L, the concentration that was midway between the baseline and the concentration that gave maximal effect (Binder et al., 1997). However, DON can depress microbial protein production (Danicke et al., 2005). As shown in swine, other co-occurring toxins such as fusaric acid may increase the negative effects of DON (Smith et al., 1997).

No DON was detected in the milk of cows provided with diets containing 66 mg/kg DON for 5 days (detection limit 1 pg/kg; Côté et al., 1986). The metabolite DOM-1 in the milk ranged from undetectable levels to 26 pg/kg, varying greatly by cow, but was no longer detectable 16 hours after the last DON feeding. At DON ingestion levels of 18.8 to 60.8 mg/d, milk concentrations ranged from 0.11 to 0.22 pg/kg for DON and 1.5 to 2.9 pg/kg for DOM-1. For DON, < 0.02 percent of intake was found in milk (Seeling et al., 2006). The FDA advisory level for DON that apparently applies to dairy cattle is 5 mg/kg in grains and grain by-products with the added recommendation that these ingredients not exceed 40 percent of the diet (FDA, 2005a); DON values are on an 88 percent DM basis (National Grain and Feed Association, 2011).

Fumonisin

Fumonisin are mycotoxins produced in the field or in storage by a variety of *Fusarium* spp., including *Fusarium verticillioides* and *Fusarium proliferatum* (Voss et al., 2007). The toxins are found predominantly in corn grain and associated products (FDA, 2001). However, they have also been found in small grain products (Batatinha et al., 2007) and can be found in sorghum and millet (Nelson et al., 1991). High levels of contamination have been associated with hot and dry weather followed by periods of high humidity during the growing season. Many different fumonisins are produced, with FB₁ and FB₂ having the greatest concentrations under natural conditions and the greatest toxicological significance (Thiel et al., 1992). Fumonisin FB₁ can cause pulmonary edema in swine (Harrison et al., 1990) and leukoencephalomalacia in horses, and it can be hepatotoxic and hepatocarcinogenic in mice (Thiel et al., 1992). It has been statistically associated with an incidence of esophageal cancer in humans (Thiel et al., 1992), but it has not yet been proven to cause disease in humans (Voss et al., 2007). Ruminants are largely tolerant of fumonisins apparently because they are minimally absorbed. In beef steers, more than 80 percent of ingested FB₁ and FB₂ was excreted in the feces (Smith and Thakur, 1996). The estimated carryover rate into milk of fumonisin B₁ is 0 to 0.05 percent of intake (Finks-Gremmel, 2008). The FDA (2005a) guidance on fumonisins in animal feeds recommends maximum levels of fumonisins FB₁ + FB₂ + FB₃ of 15 mg/kg in the total rations for bulls, lactating dairy cows, and breeding stock and 30 mg/kg for cattle that are >3 months old and fed for slaughter; fumonisin values are on a dry weight basis (National Grain and Feed Association, 2011).

Ochratoxin

Ochratoxin A is produced by *Aspergillus* and *Penicillium* spp. (Merck, 2010c). It may be found in small grains, corn, and other feeds that have molded (Pohland et al., 1992). It is extensively degraded to nontoxic ochratoxin a and phenylalanine in ruminal, reticular, and omasal digesta samples (Hult et al., 1976), leaving functioning ruminants relatively resistant to all but very high (12 mg ochratoxin A/kg BW) pulse doses (Ribelin et al., 1978). Calves with functioning rumens may suffer no ill effects of ochratoxin A, but milk-fed calves that received a pulse dose of >1 mg/kg of BW died (Sreemannarayana et al., 1988). However, milk-fed calves would probably not consume that much because they consume limited amounts of grain. Studies in Europe have detected ochratoxin A in milk (range of 5 to 40 pg/kg) in 1 to 15 percent of the milk samples tested, depending on the study (Battacone et al., 2010). No tolerances or guidance have been established for levels of ochratoxin in animal feeds (FDA, 2005a).

Patulin

Patulin has received limited study for its effects on dairy cattle. It is produced by several of the *Aspergillus*, *Penicillium*, and *Byssoschlamys* species of molds (Puel et al., 2010). Although there is no definitive information on its health effects in ruminants, it alters rumen fermentation in vitro, depressing microbial protein production, substrate digestibility, and volatile fatty acid production (Tapia et al., 2005). There are no tolerances or guidance established for its content in feeds, or information on its carryover into milk. As a food contaminant, it is more commonly associated with apples and their products (FDA, 2005b).

Zearalenone

Zearalenone is a mycotoxin with estrogenic effects produced by *Fusarium* spp. Feeds that contain zearalenone also typically contain deoxyvalenol, which is another *Fusarium* spp.-produced mycotoxin (Mirocha et al., 1976). Zearalenone has been detected in moldy grains, in silages (Kalac, 2011), and in grass and legume pastures at levels that could affect animal performance (Reed et al., 2004). In pastures, the concentrations were independent of mean annual rainfall, date of sampling, pasture height, and pasture age. Production of zearalenone is favored by high humidity and low temperatures during the growing season. More than 90 percent of zearalenone is converted to zearalenol by rumen microbes, with approximately twice as much a-zearalenol produced compared to P-zearalenol; protozoa were more active in this conversion than were bacteria (Kiessling et al., 1984). In rat uterus bioassays, a-zearalenol is three times more estrogenic than zearalenone, and P-zearalenol is equal in activity to the parent compound (Hagler et al., 1979). The mycotoxin and its degradation products can bind to estradiol-17p receptors with clinical effects indistinguishable from excessive estrogen administration (Merck, 2010c). Dietary concentrations of more than 10 and 20 mg/kg may cause reproductive dysfunction in dairy heifers and mature cows, respectively; young male cattle may become infertile (Merck, 2010c). The concentrations of zearalenone and a- and P-zearalenol in milk from cows consuming 238 to 1,125 pg zearalenone/d were below detection limits (1,3, and 1 pg/kg, respectively) (Seeling et al., 2005). Mirocha et al. (1981) reported a 0.7 percent carryover of zearalenone and its metabolites into milk. No tolerances or guidance have been established by the FDA for zearalenone in animal feeds (FDA, 2005a).

MICROBES AS TOXIC AGENTS IN FEED OR WATER

Botulism, *Clostridium botulinum*

The neurotoxin producing *Clostridium botulinum* is a widely distributed anaerobic spore-forming rod found in soils, sediments in streams and bodies of water, and the intestinal

tracts of animals (FDA, 2012b). The neurotoxin is produced during the growth of the organism. Of the seven types of botulinum toxin, types A, B, E, and F cause human botulism; types C and D cause botulism in animals (FDA, 2012b), but botulism in cattle may also be caused by types A and B (Lindstrom et al., 2010). The signs of botulism include muscle paralysis and weakness, including progressive motor paralysis, problems in chewing and swallowing, inability to rise, and death (Merck, 2010b). Signs of poisoning may occur within 24 hours to several weeks after ingestion (Myllykoski et al., 2009). Immunization of cattle against type C and D botulism has been used as a preventive measure (Merck, 2010b).

Intoxication of humans or animals most commonly occurs when they consume foods or feeds contaminated with the toxin but may also occur if *C. botulinum* grows and produces toxin in the intestinal tract. With cattle, contaminated feeds are the most common cause. Forages contaminated with decayed carcasses of animals accidentally incorporated during forage harvest are a known source of toxin (Myllykoski et al., 2009). Because *Clostridia* and their spores are ubiquitous, they do contaminate forages, but production of the botulinum toxin is dependent on storage conditions. The higher the water content of the forage and the higher the pH, the greater potential for toxin production (Notermans et al., 1979). Grass or small grain silages that are not well wilted and contain insufficient sugars to support the acid production needed for preservation and acidification are at higher risk to contain the toxin. Spoiled or toxin-contaminated feeds must not be fed to prevent botulism poisoning.

It appears unlikely that botulinum toxin is transmitted via milk from cattle suffering from botulism (Lindstrom et al., 2010). A greater concern is the contamination of milk products with *C. botulinum* spores, which then produce botulinum toxin under storage conditions. The small number of reported outbreaks associated with milk products suggests a low incidence of spores in milk or the presence of competing bacteria that reduce the potential for clostridial growth (Lindstrom et al., 2010). However, attention to appropriate thermal processing, fermentation, maintenance of appropriate storage temperature, and avoiding contamination during and after processing of dairy products are critical to food safety, as it is for any processed food products. Standard pasteurization conditions of 72°C for 15 seconds inactivate at least 99.99 percent of type A and B botulinum toxins added to milk (Weingart et al., 2010).

Cryptosporidiosis, Cryptosporidium parvum

Cryptosporidium parvum, the cause of cryptosporidiosis, is an obligate, intracellular protozoan parasite that is transmitted by ingestion of oocysts shed in feces of infected animals (FDA, 2012c). Contamination of water or feed is a common route of infection, although transmission of oocysts from animal to animal, as well as indirectly by human transmission, is possible (Merck, 2010a). Cryptosporidiosis

is mostly a disease of young calves and can be found in 48 percent (Garber et al., 1994) to 70 percent (Merck, 2010a) of calves 1 to 3 weeks of age. The clinical symptoms of cryptosporidiosis in calves include transient, mild to severe diarrhea, usually with complete recovery. Treatments for cryptosporidiosis in calves are not currently available in the United States (Merck, 2010a). Reducing the incidence of infection through avoidance of transmitting oocytes between calves and through contaminated materials is recommended. Some but not all disinfectants or disinfecting methods are effective in reducing oocyst infectivity. *Cryptosporidium parvum* oocysts are resistant to disinfection with chlorine (Shields et al., 2008). Effective disinfectants include 5 percent ammonia, formalin, freeze-drying, ammonium hydroxide, hydrogen peroxide, chlorine dioxide, and 10 percent formol saline (Merck, 2010a); caution is urged in determining which of these are appropriate and approved for use under farm and feeding conditions. Temperatures less than 0°C and greater than 65°C destroy infectivity of oocytes; allowing calf feces to dry reduces infectivity, and allowing cleaned calf rearinghouses to dry for several weeks before reuse is recommended (Merck, 2010a).

People coming in direct contact with feces from infected calves or ingesting oocyst-contaminated soil, water, or food may become infected. For immunocompetent individuals, cryptosporidiosis may present as diarrhea and abdominal cramps that last for 1 to 10 days (Current et al., 1983). After the diarrhea resolves, individuals can excrete oocysts for the next several months (FDA, 2012c). Immunodeficient individuals are at a greater risk of severe health impact and may suffer from prolonged and severe diarrhea (Current et al., 1983). Cryptosporidiosis can be a waterborne disease because of outbreaks associated with drinking water and recreational water (Painter et al., 2016), but theoretically, any food touched by an infected food handler or contaminated with an environmental source of oocysts (contaminated fecal material, contaminated water supplies) can infect people consuming those products (FDA, 2012c). Produce (Painter et al., 2013), apple cider (Blackburn et al., 2006), and unpasteurized milk (Harper et al., 2002; Rosenthal et al., 2015) have been implicated in cryptosporidiosis cases. Pasteurization appears to destroy infectivity of *Cryptosporidium* oocysts. *C. parvum* oocysts suspended in water or whole milk and pasteurized at 71.7°C for 5, 10, or 15 seconds were found not to be infective (0 of 177 mice), whereas all mice (80 of 80) became infected when dosed with nonpasteurized oocytes in water or whole milk; dose was 10,000 oocysts (Harp et al., 1996).

Cyanobacteria

Cyanobacteria are a diverse group of photosynthetic bacteria, including the toxigenic genera *Microcystis*, *Anabaena*, and *Planktothrix* (Wiegand and Pflugmacher, 2005). Their growth can be accelerated with increased inputs of nutrients

such as phosphorus and N such that they can form blooms in surface water (Blaha et al., 2009). Rather than being infectious, these microbes can produce an array of toxins that include hepatotoxins, neurotoxins, cytotoxins, dermatotoxins, and irritant toxins (Wiegand and Pflugmacher, 2005). Ingestion of water from a source supporting a cyanobacteria! bloom can result in acute poisoning. Death may occur in a few hours to a few days. Signs of poisoning may include coma, muscle tremors, paddling, and labored breathing; hemorrhage and necrosis of the liver occur (Merck, 2010c). Surviving animals may recover but may show signs of photosensitization and should be housed out of direct sunlight and offered ample uncontaminated water and good-quality feed (Merck, 2010c). Tests with dairy cattle in which lethal cell concentrations of the cyanobacterium *Microcystis aeruginosa* were provided in the water (Orr et al., 2001) or in which the cyanobacterial toxin microcystin-LR, a cyclic heptapeptide, was dosed daily (Feitz et al., 2002) showed less than 0.2 ng microcystin-LR/L of milk. Interference by milk proteins made it difficult to measure the toxin at lower concentrations (Orr et al., 2001). The World Health Organization (WHO) suggests a tolerable daily limit of 0.04 µg microcystin-LR/kg of BW for humans (WHO, 2003).

Enterohemorrhagic Escherichia coli 0157:H7

Certain serotypes of *Escherichia coli* such as *E. coli* O157:H7 produce Shiga toxins and cause severe illness in humans (Riley et al., 1983). In humans, the symptoms of infection include bloody diarrhea and, in some cases, hemolytic uremic syndrome, which can result in acute renal failure (Pennington, 2010). *E. coli* O157:H7 is not pathogenic in cattle, but dairy cattle and calves can carry the organism with a prevalence ranging from 0.4 to 48.8 percent (Pennington, 2010). The contamination of foods with feces from infected animals and failure to destroy the organisms through cooking or pasteurization are the basis for outbreaks of this foodborne illness; pasteurization of milk is effective against transmission (Pennington, 2010). Feed and feed ingredients in which *E. coli* O157:H7 is detected are considered adulterated and not allowable as animal feed (FDA, 2005a).

Listeria monocytogenes

Listeria monocytogenes is a Gram-positive, aerobic or facultative anaerobic bacteria found in soil, moist environments, and decaying vegetation (FDA, 2012d). Silage, particularly when affected by aerobic spoilage, is a potential source of *Listeria* spp. contamination, as is bovine feces (Kalac, 2011). Silage pH <4.9 (Pauly and Tham, 2003) is associated with a low presence of *Listeria* spp., but some may survive at that pH for 30 days but typically not for more than 90 days. The incidence of *Listeria* spp. increases with increasing pH (Perry and Donnelly, 1990). Infection causes abortion, perinatal mortality, encephalitis, or meningitis in

ruminants (Merck, 2010b). The encephalitis may present as “circling disease,” and animals may be anorectic, depressed, and disoriented (Merck, 2010b). With removal of offending feedstuffs and sanitation to prevent fecal contamination of feed and water to reduce the risk of reinfection, and with treatment of affected animals, the recovery rate in cattle approaches 50 percent (Merck, 2010b).

In addition to animal health concerns, *L. monocytogenes* contamination of foods is a human health concern. Most seriously affecting immunocompromised individuals, pregnant women, and the elderly, the fatality rate has been reported as 20 to 25 percent, and infection may cause abortion or stillbirth (Hitchins and Whiting, 2001). Otherwise healthy individuals may have no or mild symptoms (FDA, 2012d). Among the many foods associated with *L. monocytogenes* infection are raw milk, inadequately pasteurized milk, and soft cheeses, in addition to raw vegetables, meat and meat products, and raw or smoked fish (FDA, 2012d). Control of *L. monocytogenes* in foods can be difficult because of its ability to grow slowly at refrigeration temperatures and survival of freezing and use of salt as a food preservative (Hitchins and Whiting, 2001). Foods that are pasteurized “are not reasonably likely to contain *L. monocytogenes*” (FDA, 2008).

Salmonella spp.

Salmonella spp., particularly subspecies of *S. enterica*, are non-spore-forming Gram-negative bacteria that can cause enteric disease in cattle (Merck, 2010a). The serotypes Typhimurium, Dublin, and Newport are those likely to affect cattle (Merck, 2010a). *Salmonella* infection may be endemic to a herd (Merck, 2010a) or transmitted into herds by animals or humans bringing contaminated materials, introduction of infected animals into herds, or transmission by birds (McDonough et al., 1999) and rodents (Merck, 2010a). Contamination of feed and water by feces from infected animals is a primary route of transmission. Affected adult animals or those more than 1 week of age may show acute enteritis, with fever and diarrhea, whereas newborn calves may suffer depression, fever, pneumonia, and death (Merck, 2010a).

S. enterica is also the salmonella of greatest public health concern (FDA, 2012e) and has caused severe disease in people who drank infected raw milk (McDonough et al., 1999). Commercially sold feed and feed ingredients in which *Salmonella* are detected are considered adulterated and are not allowable as animal feed (FDA, 2005a). Recommended pasteurization procedures for milk kill *S. enterica* (Marth, 1969; FDA, 2015a).

PRIONS

Prions are small proteinaceous infectious particles that can cause transmissible spongiform encephalopathies (TSEs), which are degenerative disorders of the central

nervous system (Prusiner, 1982). The TSEs include kuru and Creutzfeldt-Jakob disease (CJD) in humans, scrapie in sheep and goats, chronic wasting disease in deer and elk, transmissible encephalopathy in mink, and bovine spongiform encephalopathy (BSE, "mad cow disease") in cattle. Unlike other infectious agents, prions are small segments of protein and do not contain genetic material made of nucleic acids.

Despite the name itself indicating an infectious agent, noninfectious forms of prions have been identified in animals (Collinge et al., 1996), although the normal function of prions is not completely understood. Those found in fungi can alter cellular processes and are "epigenetic determinants," in that they may be modified by molecular events other than changes in an organism's genetic code (Tuite and Serio, 2010). Prions, like most proteins, have a specific three-dimensional structure that is essential to them performing their function. The disease-causing prions are misfolded and can induce normally folded prions to become misfolded, as well. The accumulation of misfolded prions leads to TSE (FDA, 2012f).

The variant CJD (vCJD) in humans is caused by the same prion strain that causes BSE in cattle (Collinge et al., 1996; Hill et al., 1997) and is acquired by consumption of meat contaminated with the abnormal prions (FDA, 2012f). Bovine tissues with the highest risks of carrying the disease-causing prions are skull, brain, the nerves attached to the brain, eyes, tonsils, spinal cord, nerves attached to the spinal cord, and the distal small intestinal ileum (FDA, 2012f). Milk and bovine meat free of central nervous system tissue have, to date, shown no infectivity (FDA, 2012f). Other TSEs exist in animals, but these are not known to be transmitted to humans (FDA, 2012f). The most effective means of preventing infection of humans with vCJD is prevention of infection of cattle with BSE—hence, the ban since 1997 on feeding most ruminant protein products to ruminants with the exceptions of blood products, gelatin, and tallow as described in the Code of Federal Regulations (FDA, 2015e). Subsequently, in 2008, all cattle tissues at highest risk of carrying infective prions were banned for use in all animal feed (FDA, 2015f). As of 2018, six cases of BSE in cattle have been detected in the United States since 2003.

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